Industrial Policy in Motion: Lessons from a Century of Shipbuilding

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Abstract

Industrial policy has been used throughout history in some form or other by most countries. Yet, it remains one of the most contentious issues among policy makers and economists alike. In part, this is because the empirical evidence on whether and how it should be implemented remains slim. Scant data on government subsidies, conflicting theoretical arguments, and the need to account for governments' short and long-run objectives, render research particularly challenging. In this article, we summarize recent developments that rely on estimating industry equilibrium models. We illustrate this approach using the global shipbuilding industry as a prototypical example of an industry targeted by industrial policy, especially in periods of heavy industrialization.

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Industrial policy (IP) refers to a government agenda that shapes industry structure by promoting certain industries or sectors. Although casual observation suggests that IP can boost sectoral growth, researchers and policymakers have not yet mastered predicting or evaluating the efficacy of different types of government interventions, nor how to measure its overall welfare effects. What are the short and long run welfare implications of such policies for the local and global economy? How should industrial policy actually be done? Systematic and convincing empirical evidence that addresses these questions is disproportionally slim. In our research, we focus on one particular example of IP, which serves as a revealing case studies: the shipbuilding sector.

Shipbuilding has always held an allure for governments, many of which have targeted shipbuilding over the centuries, for various reasons: the connection between trade, shipping, and shipbuilding; the development of heavy manufacturing as a strategy for promoting economic growth; employment; national security and military considerations; and the desire for national prestige (or as Stråth [1987] puts it, "pride and machismo").

The history of shipbuilding is as tumultuous as the seas themselves. Figure 1 gives us a glimpse of its modern history and illustrates how it is defined by the succession of different countries as the world's dominant shipbuilding nation. The United Kingdom holds the lion's share of the industry for the better part of the 19th and 20th centuries, fending off competition from other Western European economies (mainly Germany and Scandinavia) at times. After World War II, it is swiftly overtaken by Japan, which prevails as a world leader until the 1980s, when South Korea enters and quickly dominates the global market. In the mid-200s China takes its turn.

Why have certain countries dominated global shipbuilding? Was it the outcome of central planning? And if so, why do governments target this sector?

The UK's leading position in shipbuilding, which lasted for decades up until the 1950s, initially stemmed from having access to cheap iron and steel in the late 1800s [Hanlon, 2020] and was subsequently related to its strong maritime trade: UK trade flows dominated global trade and the British Empire required ships to execute this trade volume with its colonies and other trading partners.

During World War I and II, the United States achieved striking production levels of ships: motivated by its military needs, massive shipbuilding capacity was built and later dismantled in a very short timespan [Thornton and Thompson, 2001, Thompson, 2001]. Beyond these two incidents, the US was never globally competitive as a ship producer. This is in spite of the fact that the US had long been a proponent of industrial policy in shipbuilding (e.g. Jones Act among others). Once the high wartime demand for ships had evaporated, US shipyards, with their much higher costs, were unable to compete commercially with European and Japanese yards [Stopford, 2009].

After World War II, in its efforts to rebuild its industrial base, Japan developed national programs for its shipbuilding industry, alongside several other heavy industries, such as steel and coal. An island nation, with a very strong maritime tradition, Japan swiftly became the world's dominant ship producer. Through a series of interventions, including subsidized financing, export credits and protectionist measures, a number of Japanese conglomerates active in shipbuilding (as well as several other sectors) became global leaders.

By the 1980's, however, Japan was losing ground to South Korea. South Korean shipbuilding,

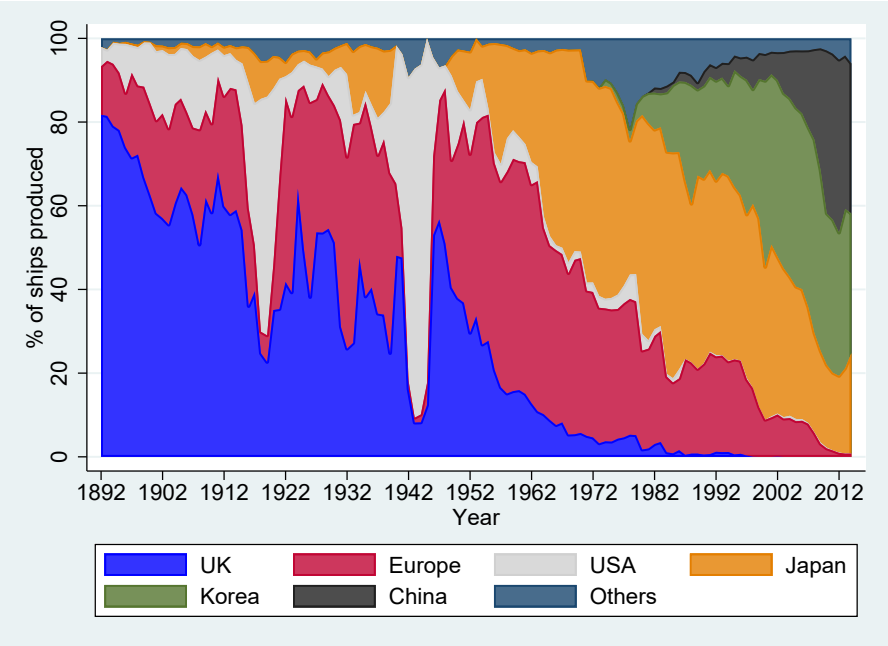

Figure 1: This figure plots the share of ships produced by each country, from 1892 - 2014. Data for 1892 - 1997 was obtained from historical issues of the World Fleet Statistics published by Lloyd's Register, while the data from 1998 onwards is based on Clarksons data.

similarly to Japanese shipbuilding, grew as part of the government's large-scale push for heavy industrialization in the late 1970's [Choi and Levchenko, 2021, Lane, 2022]. Recognizing shipbuilding as a strategic industry, the government provided support primarily in the form of favorable financing (such as low-interest loans and government debt guarantees), as well as direct investment in shipbuilding facilities. However, unlike its predecessors, South Korea did not have a maritime tradition, nor a large national fleet. This was the first time that shipbuilding had been targeted as a primarily exporting sector.

In the 2000s, China enters the shipbuilding scene. In 2002, former Premier Zhu inspected the China State Shipbuilding Corporation (CSSC), one of the two largest shipbuilding conglomerates in China, and pointed out that "China hopes to become the world's largest shipbuilding country (in terms of output) [...] by 2015." Within a few years, China overtook Japan and South Korea to become the world's leading ship producer in terms of output. Figure 2, Panel A shows the rise in China's global market share of shipbuilding by plotting China's total shipbuilding output as a share of global output. China's national and local governments provided numerous subsidies for shipbuilding, which we classify into three groups. First, below-market-rate land prices along the coastal regions, in combination with simplified licensing procedures, acted as "entry subsidies" that incentivized the creation of new shipyards. As shown in Panel (b) of Figure 2, between 2006 and 2008, the annual construction of new shipyards in China exceeded 30 new shipyards per year; in comparison, during the same time period, Japan and South Korea averaged only about one new shipyard per year each.

Second, regional governments set up dedicated banks to provide shipyards with "investment subsidies" in the form of favorable financing, including low-interest long-term loans (a common IP tool, as

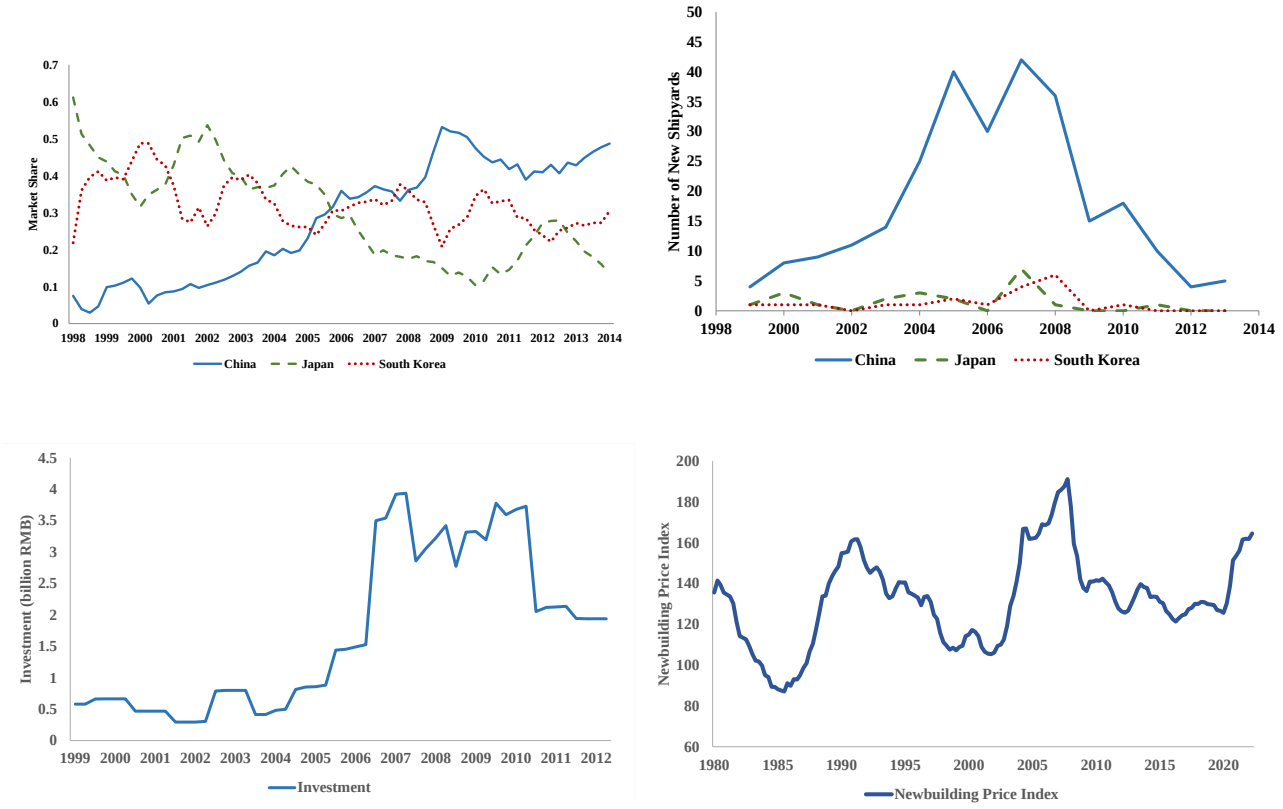

Figure 2: Figures (a) - (c) illustrate the rapid expansion of China's shipbuilding industry. (a) Source: Clarkson Research. Market shares by country are computed from quarterly ship orders. (b) Source: Clarkson Research. Number of new shipyards annually and by country. (c) Source: China's National Bureau of Statistics. Industry aggregate quarterly investment by Chinese shipyards in billions of 2000 RMB. (d) Source: Clarksons Research. Quarterly newbuilding price index (=100 in January 1988).

illustrated also by the programs in Japan and South Korea) and preferential tax policies. China's rise in total capital invested in shipyards is illustrated in Panel (c) of Figure 2. Third, China's government also employed "production subsidies" of various forms, such as subsidized material inputs, export credits, and buyer financing. The government-buttressed domestic steel industry provided cheap steel, which is an important input for shipbuilding. Export credits and buyer financing by government-directed banks made the new and unfamiliar Chinese shipyards more attractive to global buyers.

The combination of these policies was followed by a sharp expansion in China's shipbuilding production, market share, and capital accumulation. China's market share grew from 14 percent in 2003 to 53 percent by 2009, while Japan shrunk from 32 percent to 10 percent and South Korea from 42 percent to 32 percent. Then came the Great Recession of 2008-09, which drove the global shipping industry to a historic bust. The large number of new Chinese shipyards exacerbated low capacity utilization and contributed to plummeting ship prices around the world. The effectiveness of China's IP was questioned. In response to the crisis and in an effort to promote industry consolidation, the government unveiled the "2009 Plan on Adjusting and Revitalizing the Shipbuilding Industry" that resulted in an immediate moratorium on entry and subsequently shifted support towards only selected

firms in an issued "White List."

The history of shipbuilding presents puzzles and leaves us with open questions: Why have governments subsidized shipbuilding throughout history? Was industrial policy successful? Although several national programs, such as Japan's or South Korea's heavy industry drives, seem to be behind tremendous sectoral growth, others, like the US's, have seemingly failed. Moreover, even when growth is achieved, what is its welfare impact, both domestically and internationally, as well as the efficacy of different types of government interventions? Finally, how large was government support? Even this apparently simple question is surprisingly difficult to answer due to an "obscure jungle of subsidies" [Stråth, 1987]. Indeed, any methodology for estimating effects of industrial policy must grapple with a basis data challenge: that is, government subsidies to industries are notoriously difficult to detect and measure. Indeed, "systematic data are non-existent; reliable sources of information are scarce and mostly incomplete [...] because governments do not systematically provide the information" [WTO, 2006]. Thus, researchers and policymakers end up relying mostly on data reported by governments.

In recent research (Kalouptsidi [2018] and Barwick et al. [2024]) we study the impact of China's 21st-century shipbuilding program on industry evolution and global welfare, addressing the above challenges. To our knowledge, this work is the first attempt at evaluating quantitatively IP in shipbuilding globally and among the first papers employing the structural industrial organization methodology to understand the welfare implications and effective design of IP more generally.

Our methodology amounts to providing a quantitative framework for the shipbuilding industry, that explains and predicts how global shipyards optimally choose their production, investment, and whether to begin or end operations. We use this quantitative model to measure hidden subsidies, and evaluate their welfare impact globally.

We find that China provided $23 billion in production subsidies between 2006 and 2013. This finding is driven by the cost function obtained from this analysis, which exhibits a significant drop for Chinese producers equal to about 13-20 percent of the cost per ship. Simply put, Chinese shipbuilding firms were "over"-producing after 2006 compared to our prediction of output without subsidies. Altogether, China provided $91 billion in subsidies along all three margins—production, entry, and investment—between 2006 and 2013. Notably, entry subsidies were 69 percent of total subsidies.

Our analysis suggests that China's IP in support of shipbuilding boosted China's domestic investment in shipbuilding by 140 percent, and more than doubled the entry rate. It also depressed exit. Overall, IP raised China's world market share in shipbuilding by more than 40 percent.

Calculating whether this increase in sectoral output should be counted as an increase in welfare is a more delicate question. First, 70 percent of China's output expansion occurred via business stealing from rival countries. There is evidence (backed by our cost estimates) that Chinese shipyards are less efficient than their Japanese and South Korean counterparts; thus, the transfer of shipbuilding to China constitutes a misallocation of global resources. Second, China's IP for shipbuilding led to considerable declines in ship prices. Third and most importantly, although China's shipbuilding subsidies were highly effective at achieving output growth and market share expansion, we find that they were largely unsuccessful in terms of (domestic) welfare measures. In the long run, the gross return rate of the adopted policy mix, as measured by the increase in lifetime profits of domestic firms divided by total

subsidies, is only 18 percent, meaning that for every $1 the government spends, it gets back 18 cents in profitability. In other words, the net return when incorporating the cost to the government was a negative 82 percent, with entry subsidies explaining a lion's share of the negative return: entry subsidies are wasteful—even by the revenue metric—and lead to increased industry fragmentation and idleness. Entry subsidies attract small and inefficient firms.

We also explore whether the policy could have been designed more effectively. For instance, a counter-cyclical policy would outperform the pro-cyclical policy that was adopted by a large margin. Moreover, if an "optimal White List" is formed—that is, the most productive firms are chosen for subsidies—the gross rate of return would be much higher.

Our results highlight why industrial policies have worked better for some countries. In East Asian countries where IP was often considered successful, the policy support was often conditioned on firm performance. In contrast, in Latin America where industrial policies often aimed at import-substitution, no mechanisms existed to weed out non-performing beneficiaries [Rodrik, 2009]. In China's modern-day IP in the shipbuilding industry, the policy's return was low in earlier years when output expansion was primarily fueled by the entry of inefficient firms but increased over time as the government relied on "performance-based" criteria via its White List. Such targeted IP design can be substantially more successful than open-ended policies that benefit all firms.

So why did China adopt this program if the return was as low? Interestingly, we find that standard arguments for IP do not seem to apply especially well in our setting. Instead, our analyses point to two alternative potential rationales. First, as China became the world's biggest exporter and a close second largest importer during our sample period, transport cost reductions from increased shipbuilding and reduced shipping costs can lead to substantial increases in its trade volume. Our estimates suggest that China's IP policy expanded global shipping fleet, reduced freight rates, and raised China's annual trade volume by 5 percent ($144 billion) between 2006 and 2013. This increase in trade was large relative to the size of the subsidies (which averaged $11.3 billion annually between 2006 and 2013). Of course, "more trade" does not translate directly into economic well-being, but the relative magnitudes are suggestive. Second, as shown in Figure 3 both military and commercial deliveries at shipyards that produce military ships experienced a multi-fold increase, although military production appeared to have accelerated after the financial crisis and continued to increase throughout the sample period, providing suggestive evidence that China's supportive policy might have benefited its military production as well.

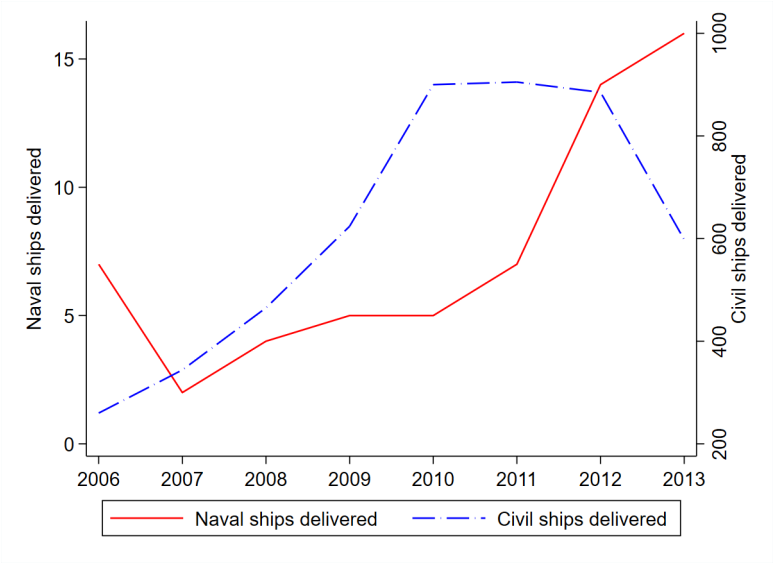

Figure 3: Military Ship Production. This figure plots the number of commercial ships and naval ships delivered by Chinese shipyards over time. Source: Clarksons and IHS Jane's.

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